

William Marshall

Compile: `g++ task_scheduler.cc -o schedule`

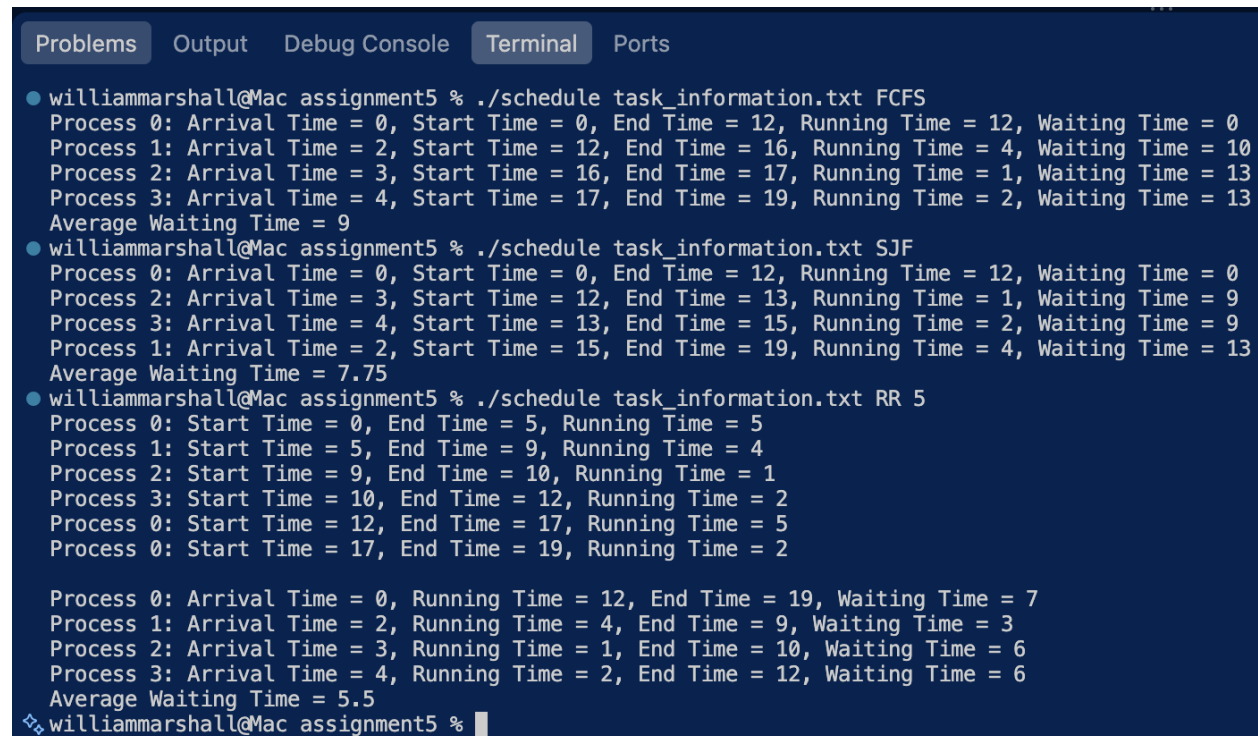
Run: `./schedule <input_file> <schedule_type>`

Schedule type is FCFS, SJF, or RR followed by an integer

Note: The assignment did not ask for all the edge cases for the inputs, so I didn't do all the checks that I could have to make run safer.

Test Cases: `task_information.txt`, `task_information_1.txt`, `task_information_2.txt`

Test Case 1:



```
Problems Output Debug Console Terminal Ports
● williammarshall@Mac assignment5 % ./schedule task_information.txt FCFS
Process 0: Arrival Time = 0, Start Time = 0, End Time = 12, Running Time = 12, Waiting Time = 0
Process 1: Arrival Time = 2, Start Time = 12, End Time = 16, Running Time = 4, Waiting Time = 10
Process 2: Arrival Time = 3, Start Time = 16, End Time = 17, Running Time = 1, Waiting Time = 13
Process 3: Arrival Time = 4, Start Time = 17, End Time = 19, Running Time = 2, Waiting Time = 13
Average Waiting Time = 9
● williammarshall@Mac assignment5 % ./schedule task_information.txt SJF
Process 0: Arrival Time = 0, Start Time = 0, End Time = 12, Running Time = 12, Waiting Time = 0
Process 2: Arrival Time = 3, Start Time = 12, End Time = 13, Running Time = 1, Waiting Time = 9
Process 3: Arrival Time = 4, Start Time = 13, End Time = 15, Running Time = 2, Waiting Time = 9
Process 1: Arrival Time = 2, Start Time = 15, End Time = 19, Running Time = 4, Waiting Time = 13
Average Waiting Time = 7.75
● williammarshall@Mac assignment5 % ./schedule task_information.txt RR 5
Process 0: Start Time = 0, End Time = 5, Running Time = 5
Process 1: Start Time = 5, End Time = 9, Running Time = 4
Process 2: Start Time = 9, End Time = 10, Running Time = 1
Process 3: Start Time = 10, End Time = 12, Running Time = 2
Process 0: Start Time = 12, End Time = 17, Running Time = 5
Process 0: Start Time = 17, End Time = 19, Running Time = 2

Process 0: Arrival Time = 0, Running Time = 12, End Time = 19, Waiting Time = 7
Process 1: Arrival Time = 2, Running Time = 4, End Time = 9, Waiting Time = 3
Process 2: Arrival Time = 3, Running Time = 1, End Time = 10, Waiting Time = 6
Process 3: Arrival Time = 4, Running Time = 2, End Time = 12, Waiting Time = 6
Average Waiting Time = 5.5
williammarshall@Mac assignment5 %
```

Test Case 2:

```
● williammarshall@Mac assignment5 % ./schedule task_information_1.txt FCFS
Process 0: Arrival Time = 0, Start Time = 0, End Time = 8, Running Time = 8, Waiting Time = 0
Process 1: Arrival Time = 1, Start Time = 8, End Time = 11, Running Time = 3, Waiting Time = 7
Process 2: Arrival Time = 2, Start Time = 11, End Time = 17, Running Time = 6, Waiting Time = 9
Process 3: Arrival Time = 4, Start Time = 17, End Time = 19, Running Time = 2, Waiting Time = 13
Process 4: Arrival Time = 5, Start Time = 19, End Time = 24, Running Time = 5, Waiting Time = 14
Average Waiting Time = 8.6
● williammarshall@Mac assignment5 % ./schedule task_information_1.txt SJF
Process 0: Arrival Time = 0, Start Time = 0, End Time = 8, Running Time = 8, Waiting Time = 0
Process 3: Arrival Time = 4, Start Time = 8, End Time = 10, Running Time = 2, Waiting Time = 4
Process 1: Arrival Time = 1, Start Time = 10, End Time = 13, Running Time = 3, Waiting Time = 9
Process 4: Arrival Time = 5, Start Time = 13, End Time = 18, Running Time = 5, Waiting Time = 8
Process 2: Arrival Time = 2, Start Time = 18, End Time = 24, Running Time = 6, Waiting Time = 16
Average Waiting Time = 7.4
● williammarshall@Mac assignment5 % ./schedule task_information_1.txt RR 5
Process 0: Start Time = 0, End Time = 5, Running Time = 5
Process 1: Start Time = 5, End Time = 8, Running Time = 3
Process 2: Start Time = 8, End Time = 13, Running Time = 5
Process 3: Start Time = 13, End Time = 15, Running Time = 2
Process 4: Start Time = 15, End Time = 20, Running Time = 5
Process 0: Start Time = 20, End Time = 23, Running Time = 3
Process 2: Start Time = 23, End Time = 24, Running Time = 1

Process 0: Arrival Time = 0, Running Time = 8, End Time = 23, Waiting Time = 15
Process 1: Arrival Time = 1, Running Time = 3, End Time = 8, Waiting Time = 4
Process 2: Arrival Time = 2, Running Time = 6, End Time = 24, Waiting Time = 16
Process 3: Arrival Time = 4, Running Time = 2, End Time = 15, Waiting Time = 9
Process 4: Arrival Time = 5, Running Time = 5, End Time = 20, Waiting Time = 10
Average Waiting Time = 10.8
❖ williammarshall@Mac assignment5 %
```

Test Case 3:

```
● williammarshall@Mac assignment5 % ./schedule task_information_2.txt FCFS
Process 0: Arrival Time = 0, Start Time = 0, End Time = 4, Running Time = 4, Waiting Time = 0
Process 1: Arrival Time = 0, Start Time = 4, End Time = 11, Running Time = 7, Waiting Time = 4
Process 2: Arrival Time = 3, Start Time = 11, End Time = 13, Running Time = 2, Waiting Time = 8
Process 3: Arrival Time = 5, Start Time = 13, End Time = 14, Running Time = 1, Waiting Time = 8
Process 4: Arrival Time = 6, Start Time = 14, End Time = 17, Running Time = 3, Waiting Time = 8
Process 5: Arrival Time = 8, Start Time = 17, End Time = 22, Running Time = 5, Waiting Time = 9
Average Waiting Time = 6.16667
● williammarshall@Mac assignment5 % ./schedule task_information_2.txt SJF
Process 0: Arrival Time = 0, Start Time = 0, End Time = 4, Running Time = 4, Waiting Time = 0
Process 2: Arrival Time = 3, Start Time = 4, End Time = 6, Running Time = 2, Waiting Time = 1
Process 3: Arrival Time = 5, Start Time = 6, End Time = 7, Running Time = 1, Waiting Time = 1
Process 4: Arrival Time = 6, Start Time = 7, End Time = 10, Running Time = 3, Waiting Time = 1
Process 5: Arrival Time = 8, Start Time = 10, End Time = 15, Running Time = 5, Waiting Time = 2
Process 1: Arrival Time = 0, Start Time = 15, End Time = 22, Running Time = 7, Waiting Time = 15
Average Waiting Time = 3.33333
● williammarshall@Mac assignment5 % ./schedule task_information_2.txt RR 5
Process 0: Start Time = 0, End Time = 4, Running Time = 4
Process 1: Start Time = 4, End Time = 9, Running Time = 5
Process 2: Start Time = 9, End Time = 11, Running Time = 2
Process 3: Start Time = 11, End Time = 12, Running Time = 1
Process 4: Start Time = 12, End Time = 15, Running Time = 3
Process 5: Start Time = 15, End Time = 20, Running Time = 5
Process 1: Start Time = 20, End Time = 22, Running Time = 2

Process 0: Arrival Time = 0, Running Time = 4, End Time = 4, Waiting Time = 0
Process 1: Arrival Time = 0, Running Time = 7, End Time = 22, Waiting Time = 15
Process 2: Arrival Time = 3, Running Time = 2, End Time = 11, Waiting Time = 6
Process 3: Arrival Time = 5, Running Time = 1, End Time = 12, Waiting Time = 6
Process 4: Arrival Time = 6, Running Time = 3, End Time = 15, Waiting Time = 6
Process 5: Arrival Time = 8, Running Time = 5, End Time = 20, Waiting Time = 7
Average Waiting Time = 6.66667
❖ williammarshall@Mac assignment5 %
```